

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (classification_report, confusion_matrix,
ConfusionMatrixDisplay, roc_auc_score, roc_curve)
```

```
!test -f diabetes.csv || wget https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv
```

```
--2026-06-01 10:10:22-- https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 23873 (23K) [text/plain]
Saving to: 'diabetes.csv'
```

```
diabetes.csv      100%[=====] 23.31K  --.-KB/s    in 0.001s
```

```
2026-06-01 10:10:22 (16.9 MB/s) - 'diabetes.csv' saved [23873/23873]
```

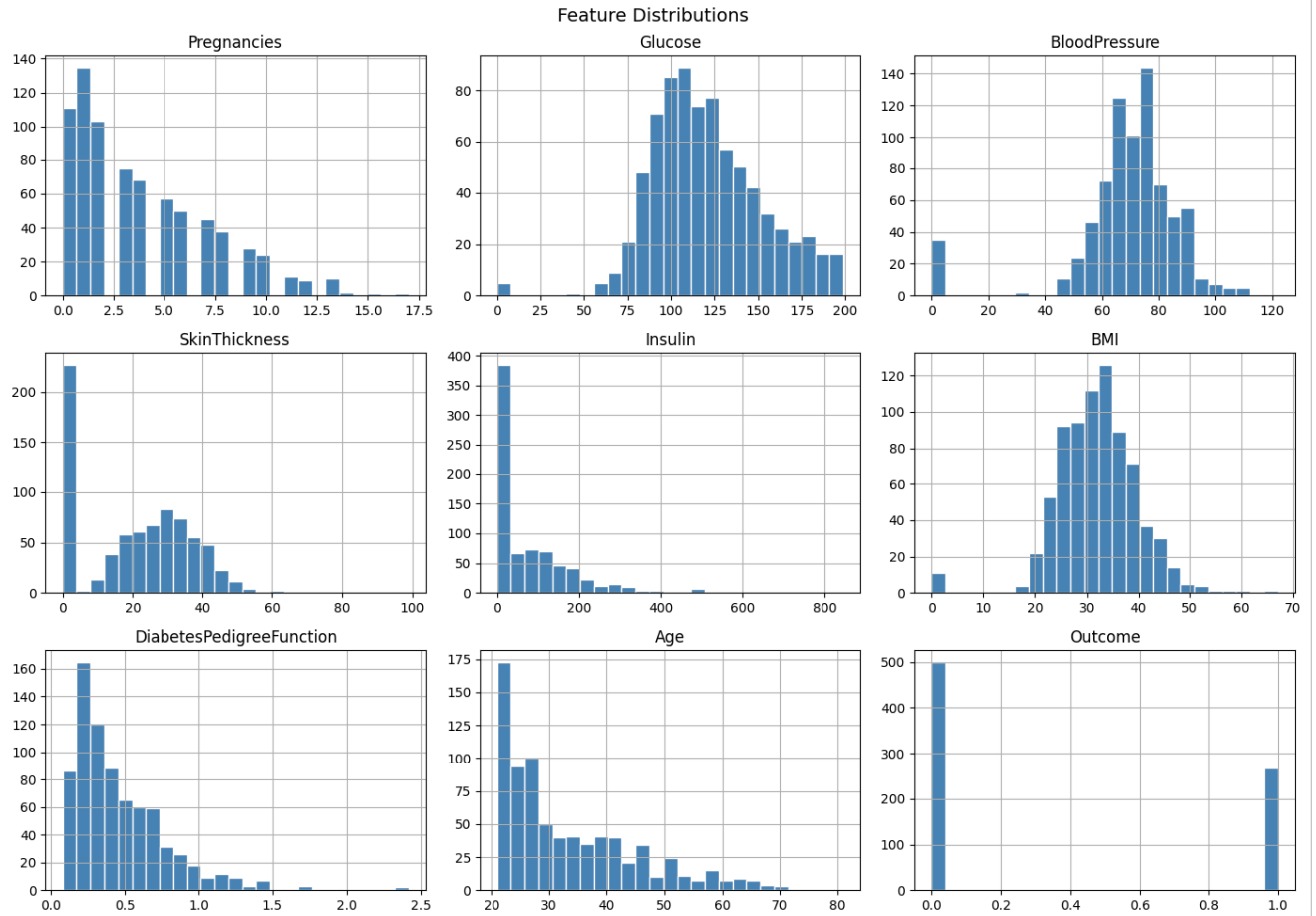
```
df = pd.read_csv('diabetes.csv')
print(df.shape)
print(df['Outcome'].value_counts())
print(df.describe().round(2))
```

```
(768, 9)
Outcome
0    500
1     268
Name: count, dtype: int64
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
count	768.00	768.00	768.00	768.00	768.00	768.00
mean	3.85	120.89	69.11	20.54	79.80	31.99
std	3.37	31.97	19.36	15.95	115.24	7.88
min	0.00	0.00	0.00	0.00	0.00	0.00
25%	1.00	99.00	62.00	0.00	0.00	27.30
50%	3.00	117.00	72.00	23.00	30.50	32.00
75%	6.00	140.25	80.00	32.00	127.25	36.60
max	17.00	199.00	122.00	99.00	846.00	67.10

	DiabetesPedigreeFunction	Age	Outcome
count	768.00	768.00	768.00
mean	0.47	33.24	0.35
std	0.33	11.76	0.48
min	0.08	21.00	0.00
25%	0.24	24.00	0.00
50%	0.37	29.00	0.00
75%	0.63	41.00	1.00
max	2.42	81.00	1.00

```
df.hist(bins=25, figsize=(14,10), color='steelblue', edgecolor='white')
plt.suptitle('Feature Distributions', fontsize=14); plt.tight_layout(); plt.show()
```



```
fix_cols = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
for col in fix_cols:
    n = (df[col] == 0).sum()
    df[col] = df[col].replace(0, df[col].median())
    print(f'{col:25s}: {n} zeros → replaced with median {df[col].median():.1f}')
```

```
Glucose           : 5 zeros → replaced with median 117.0
BloodPressure     : 35 zeros → replaced with median 72.0
SkinThickness     : 227 zeros → replaced with median 23.0
Insulin           : 374 zeros → replaced with median 31.2
BMI               : 11 zeros → replaced with median 32.0
```

```
X = df.drop('Outcome', axis=1)
y = df['Outcome']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression(max_iter=1000, random_state=42)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

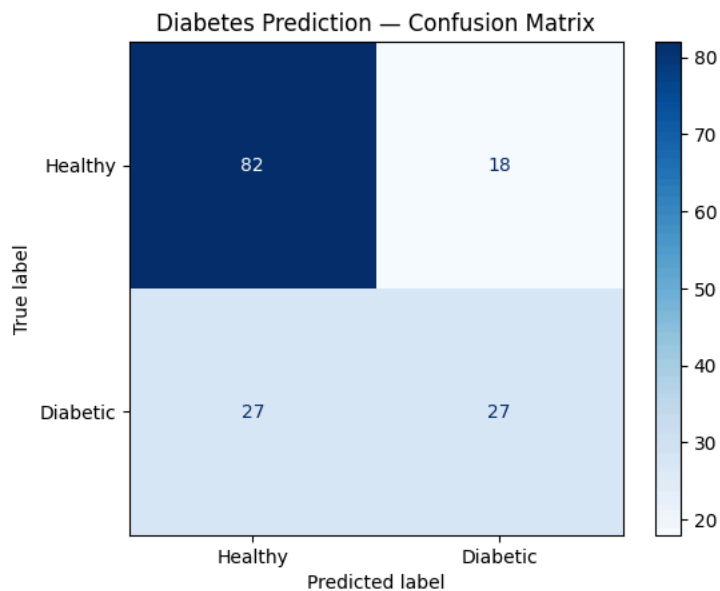
```
for name, coef in zip(X.columns, model.coef_[0]):
    print(f'{name:30s}: {coef:+.3f}')
```

```
Pregnancies      : +0.376
Glucose          : +1.209
BloodPressure    : -0.054
SkinThickness    : +0.029
Insulin          : -0.133
BMI              : +0.701
DiabetesPedigreeFunction : +0.245
Age              : +0.138
```

```
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]
print(classification_report(y_test, y_pred, target_names=['Healthy', 'Diabetic']))
```

	precision	recall	f1-score	support
Healthy	0.75	0.82	0.78	100
Diabetic	0.60	0.50	0.55	54
accuracy			0.71	154
macro avg	0.68	0.66	0.67	154
weighted avg	0.70	0.71	0.70	154

```
cm = confusion_matrix(y_test, y_pred)
ConfusionMatrixDisplay(cm, display_labels=['Healthy', 'Diabetic']).plot(cmap='Blues')
plt.title('Diabetes Prediction - Confusion Matrix'); plt.show()
```



```
fpr, tpr, _ = roc_curve(y_test, y_prob)
auc = roc_auc_score(y_test, y_prob)
plt.figure(figsize=(7.2, 3.5))
plt.plot(fpr, tpr, lw=2.5, color='steelblue', label=f'AUC = {auc:.3f}')
plt.plot([0,2],[0,2], 'k--'); plt.xlabel('FPR'); plt.ylabel('TPR')
plt.title('Diabetes Model ROC Curve'); plt.legend(); plt.tight_layout(); plt.show()
```

Diabetes Model ROC Curve



AUC = 0.815

```
new_patient = np.array([[10, 120, 120, 44, 2, 45.7, 0.627, 50]])
new_scaled = scaler.transform(new_patient) # must use same scaler
pred = model.predict(new_scaled)[0]
prob = model.predict_proba(new_scaled)[0, 1]
print(f'Prediction : {'Diabetic' if pred==1 else 'Healthy'})
print(f'Probability : {prob:.1%} chance of having diabetes')
```

```
Prediction : Diabetic
Probability : 80.2% chance of having diabetes
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
warnings.warn(
```

```
import pandas as pd
coef_df = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_[0]})
coef_df = coef_df.sort_values('Coefficient')
coef_df.plot(kind='barh', x='Feature', y='Coefficient',
color=['tomato' if c > 0 else 'steelblue' for c in coef_df['Coefficient']],
legend=False, figsize=(7, 4))
plt.axvline(x=0, color='black', linewidth=0.8)
plt.title('Feature Coefficients — Positive = increases diabetes risk')
plt.tight_layout(); plt.show()
```

